

The Ephemeris

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Sept - Nov 2016 Events

Board & General Meetings

Saturday 9/17, 10/15, 11/12, 12/10

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 9/11, 10/2, 11/06, 12/04

Solar Observing (locations differ)

Sunday 9/11, 10/2, 11/06, 12/04

Intro to the Night Sky Class

Houge Park 1Q In-Town Star Party

Friday 9/9, 10/7, 11/4, 12/02

Astronomy 101 Class

Houge Park 3Q In-Town Star Party

Friday 9/23, 10/21, 11/18, 12/16

RCDO Starry Nights Star Party

Saturday 9/24, 10/22, 11/19, 12/17

Imaging SIG Mtg

Tuesday 9/20, 10/18, 11/15, 12/20

Astro Imaging Workshops

(at Houge Park)

Saturday 10/22, 11/19

Coder's Meeting

Sunday 9/18, 10/16, 11/20, 12/18

Quick STARt (by appointment)

Saturday 10/8, Friday 12/9

Unless noted above, please refer to the SJAA Web page for specific event times, locations and possible cancellation due to weather.

Cover Photo: Orion

By SJAA Member Glenn Newell

SJAA at Lick Observatory May 2016

"What a wonderful evening tour of Lick Observatory on Friday May 13, 2016! Thank you to Bill and Susan O'Neil for coordinating the event and to Dr. Elinor Gates and Tanja Bode for their excellent presentations". **Gary Chock**

"Dr. Gates took the group for an "A" ticket "Ride the Dome" spin of the Shane telescope while the group was up on the Dome catwalk. The telescope operator showed how fast the Shane telescope can move in all directions, while the catwalk rotated around the room .

SJAA invited 40 generous members and Lick supporters to attend the Tour event at a fee of \$50 each, raising \$2,000 to support Lick Observatory. The event was held on Friday, May 13, 2016. And we got to see M3 globular Cluster in the 57' long, Great Refractor installed in 1888". **Bill O'Neil**

(see photos following page)

*Viewing Jupiter through the 36" Lick Refractor
Photo: Gary Chock*

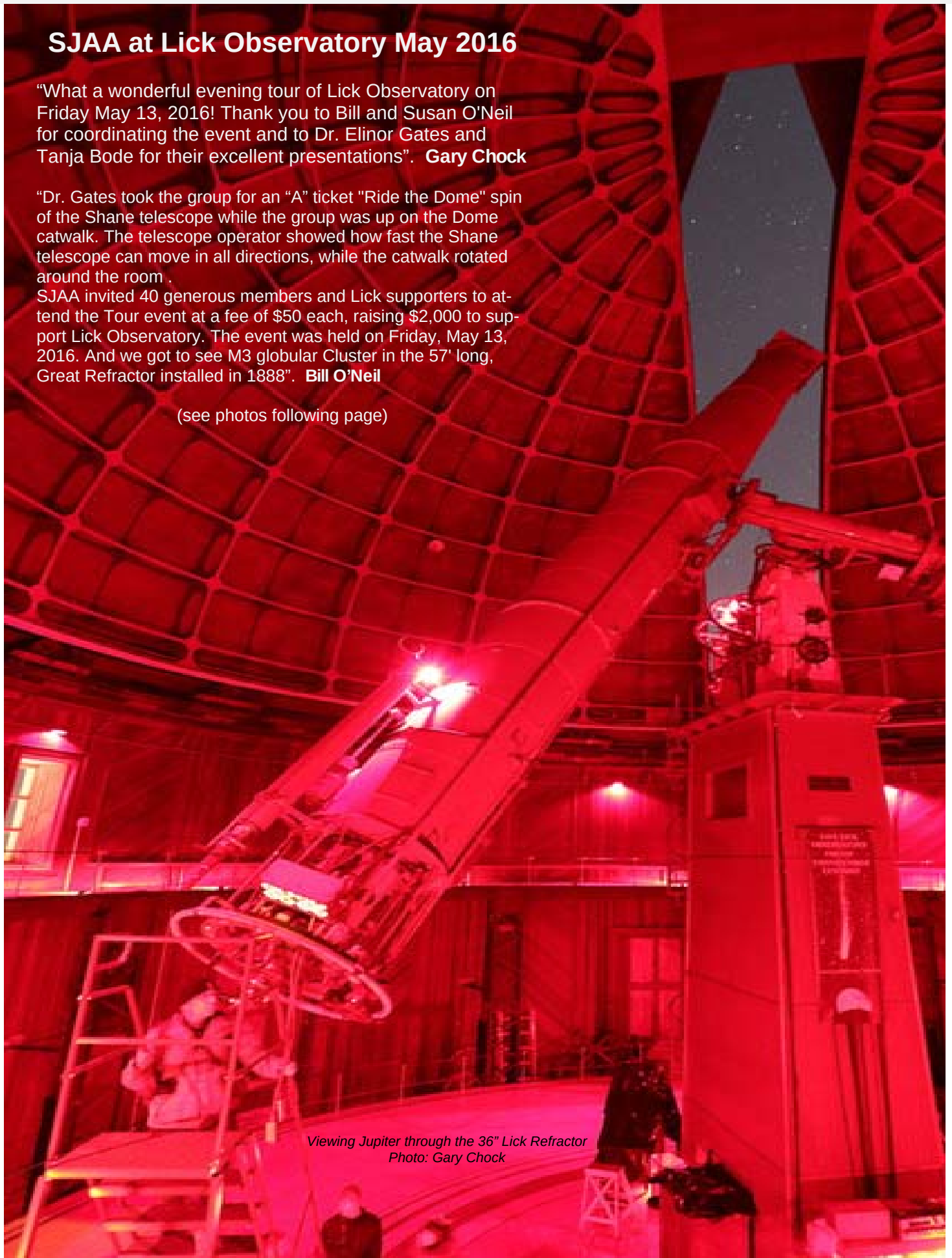
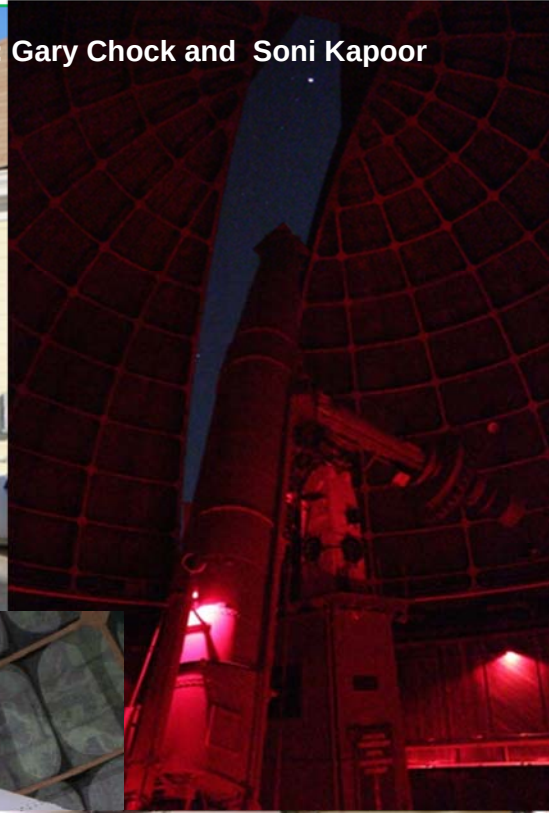


Photo credits: Gary Chock and Soni Kapoor



Dr. Elinor Gates harnessed in and maneuvered the big refractor by hand.



Shane 120" Reflector



Curiosity's 4th Anniversary on Mars

As Curiosity marks its fourth anniversary (in Earth years) since landing on Mars, the rover is working on collecting its 17th sample. While Curiosity explores Mars, gamers can join the fun via a new social media game, Mars Rover.

On their mobile devices, players drive a rover through rough Martian terrain, challenging themselves to navigate and balance the rover while earning points along the way. The game also illustrates how NASA's next Mars rover, in development for launch in 2020, will use radar to search for underground water.

For more information about how the Mars Rover game relates to exploration by NASA's Mars rovers, visit:

mars.nasa.gov/gameee-rover

Meanwhile, on Mars the real rover has driven to position for drilling into a rock target called "Marimba," to acquire rock powder for onboard laboratory analysis. The rover has begun a multi-month ascent of a mudstone geological unit as it heads toward higher and progressively younger geological evidence on Mount Sharp, including some rock types not yet explored.

The mission is examining the lower slopes of Mount Sharp, a layered mountain inside Gale Crater, to learn more about how and when ancient environmental conditions in the area evolved from fresh-water settings into conditions drier and less favorable for life. Six of the mission's 13 drilled rock-samples so far, and two of its four scooped soil samples, have been collected since the third anniversary of landing.

In its four years, Curiosity has returned more than 128,000 images and fired its laser more than 362,000 times. As of the fourth anniversary, Curiosity has driven 8.43 miles (13.57 kilometers).

Curiosity landed inside Mars' Gale Crater on Aug. 6, 2012, EDT (evening of Aug. 5, PDT), with a touchdown technique called the sky-crane maneuver. During the rover's first Earth year on Mars, the mission accomplished its main goal when it found and examined an ancient habitable environment. Researchers determined that a freshwater lake at the "Yellowknife Bay" site billions of years ago offered the chemical ingredients and energy favorable for supporting microbial life, if life has ever existed on Mars.

NASA's orbiters and rovers at Mars enable continued scientific discoveries and prepare the way for future astronauts to explore the Red Planet.



Two sizes of ripples are evident in this Dec. 13, 2015, view of a top of a Martian sand dune, from NASA's Curiosity Mars rover. Sand dunes and the smaller type of ripples also exist on Earth. The larger ripples are a type not seen on Earth nor previously recognized as a distinct type on Mars.

Credits: NASA/JPL-Caltech/MSSS

A Hubble Sky Full of Stars

Located approximately 22,000 light-years away in the constellation of Musca (The Fly), this tightly packed collection of stars — known as a globular cluster — goes by the name of NGC 4833. This NASA/ESA Hubble Space Telescope image shows the dazzling stellar group in all its glory.

NGC 4833 is one of the over 150 globular clusters known to reside within the Milky Way. These objects are thought to contain some of the oldest stars in our galaxy. Studying these ancient cosmic clusters can help astronomers to unravel how a galaxy formed and evolved, and give an idea of the galaxy's age.

Globular clusters are responsible for some of the most striking sights in the cosmos, with hundreds of thousands of stars congregating in the same region of space. Hubble has observed many of these clusters during its time in orbit around our planet, each as breathtaking as the last.



Credits: NASA.gov

Five Years Post-Launch, Juno Is at a Turning Point

Five years after departing Earth, and a month after slipping into orbit around Jupiter, NASA's Juno spacecraft reached a turning point. On July 31 at 12:41 p.m. PDT (3:41 p.m. EDT), Juno reached the farthest point in its orbit of Jupiter for the first time, known as "apojove," 5 million miles (8.1 million kilometers) from the giant planet. After that point, Jupiter's gravitational grip on Juno will cause the spacecraft to begin falling back toward the planet for another pass, this time with its scientific eyes wide open.

The spacecraft is currently executing the first of two long orbits prior to beginning its science mission. Each capture orbit is nearly two months long -- quite the wait for the mission's eager team of scientists -- but it's nothing compared to the long wait the team endured on the trek to Jupiter.

Juno launched on Aug. 5, 2011. The spacecraft took a long, looping path around the inner solar system to set up an Earth flyby, in which our planet's gravity flung the spinning probe onward toward Jupiter.

"For five years we've been focused on getting to Jupiter. Now we're there, and we're concentrating on beginning dozens of flybys of Jupiter to get the science we're after," said Scott Bolton, Juno principal investigator at Southwest Research Institute in San Antonio.

Juno arrived at Jupiter on July 4, firing its main rocket engine as planned for 35 minutes. The flawless maneuver allowed Jupiter's gravity to capture the solar powered spacecraft into the first of two 53.4-day-long orbits, referred to as capture orbits. Following the capture orbits, Juno will fire its engine once more to shorten its orbital period to 14 days and begin its science mission.

But before that happens, on Aug. 27, Juno must finish its first lap around Jupiter, with a finish line that represents the mission's closest pass over the gas giant. During the encounter, Juno will skim past Jupiter at a mere 2,600 miles (4,200 kilometers) above the cloud tops.

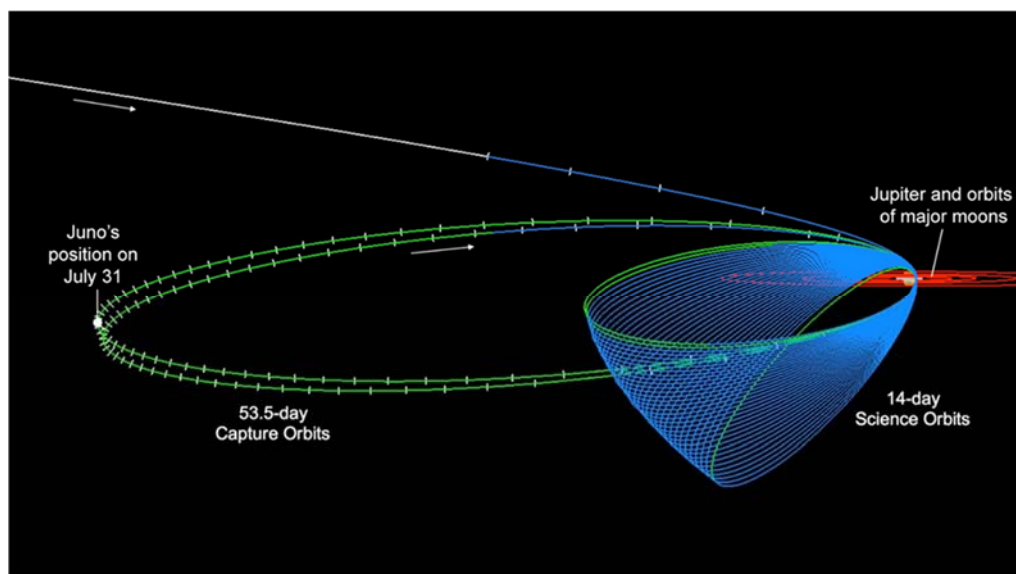
Juno's science instruments were turned off during orbit insertion, to simplify spacecraft operations during that critical maneuver. In contrast, all the instruments will be collecting data during the Aug. 27 pass, which serves as a trial run before the mission gets to work collecting the precious data it came for.

"We're in an excellent state of health, with the spacecraft and all the instruments fully checked out and ready for our first up-close look at Jupiter," said Rick Nybakken, Juno project manager at NASA's Jet Propulsion Laboratory, Pasadena, California.

With its powerful suite of science instruments, Juno will probe Jupiter's deep structure, atmospheric circulation and the high-energy physics of its magnetic environment. What Juno finds there will reveal important clues to Jupiter's formation and evolution, along with insights about how our planetary system and others are built. (Credit : nasa.gov)

The adjacent diagram shows the Juno spacecraft's orbits, including its two long, stretched-out capture orbits. The spacecraft's position on July 31 is indicated at left.

Credits: NASA/JPL-Caltech



The Autumn Sky

From Tom Piller

Source: Observer's Handbook 2016 by the Royal Astronomical Society of Canada. Copies of the handbook are available and can be ordered through SJAA typically during the month of October. Keep an eye out for the 2017 order notice from SJAA coming soon.

Planets

Mercury should be visible in the morning sky during the first half of October and then moves into the evening twilight the later half of November and through all of December. Aphelion is on November 11th and perihelion on December 25th.

Venus will continue to shine in the Western sky after sunset September through December. It will be in conjunction with Saturn on October 30th and quasi-conjunction with Antares. Aphelion is on October 31st.

Mars spends most of the month of September in Ophiuchus, moving into Sagittarius in late September and reaching perihelion October 29th. The planet then moves into Capricornus in early November and Aquarius in mid-December continuing to set mid-evening through November and early evening in December.

Jupiter vanishes into the evening twilight in early September and will be in conjunction with the Sun on September 26th. By the middle of October Jupiter reappears in the morning sky in Virgo low in the eastern dawn twilight where it remains through November. In December Jupiter rises in the east well after midnight still in Virgo.

Saturn will be low in the southwest mid-evening sky September through October setting late evening. It will be in conjunction with Venus on October 30th and quasi-conjunction with Antares. November finds Saturn vanishing into the evening twilight in the southwest and in December it is too close to the Sun to be visible.

Uranus in September rises mid-evening, retrograding in Pisces where it remains through December well placed in the evening sky for observing.

Neptune will be visible all night September through November in Aquarius. In October Neptune retrogrades in Aquarius but stays in the constellation through December setting in late evening by the end of the year.

Meteor Showers

The Orionid meteor shower peaks on Friday, October 21st

The Leonid meteor shower peaks on Thursday, November 17th.

The Geminid meteor shower peaks Wednesday, December 14th

The Ursid meteor shower peaks on Thursday, December 22nd.

Eclipses

There will be no lunar or solar eclipses visible in Northern California for the rest of 2016.

Constellations

As SJAA's Dave Ittner once explained to me one evening out at RCDO, the autumn night skies are occupied by the 'water' constellations; Aquarius (the Water Bearer), Capricornus (the sea goat), Cetus (the whale or sea monster), Delphinus (dolphin), Eridanus (river), Pisces (fish). None of the stars in these constellations are very bright to me and they are hard to pick out. A dark site makes observing them much easier (RCDO works). You might even be surprised at how dark this area of the sky (looking away from the Milky Way) can be.

Pegasus hovers over the water constellations all autumn in the south-west. Taurus and Orion make their debut in November in the east but don't really rise high enough in the night sky for good observing until December (I can hardly wait). And of course where ever Orion goes his companion Sirius is not far behind.

Deep Sky - Messier Marathon

SJAA will be hosting a 'half' Messier marathon in October to observe the Messier objects, most of which will be visible over the course of an entire night for a short period of time. Please check for the announcement in 'SJAA Meetup'.

Pinnacles DSSH June 4, 2016

From Ed Wong

Oh, what a night! That's how I would describe the SJAA Summer Deep Sky Scavenger Hunt at the Pinnacles West Side. The anticipation was high for me for this event since the winter event was cancelled due to El Nino storms. The weather was supposed to be good so I was thinking this will be good. But as with all things in life sometimes things are not so smooth. On the drive down I noticed a big cloud over the Soledad area which at first looked like the marine layer but, as I made it to Soledad I realized it was a fire and the cloud was spreading. Not good :-(as I made my way up to the Pinnacles I was thinking, I wondered if this event is going to be doomed for failure again. As I arrived, there were already a few club members there.

I started the usual setup procedures for prepping the facilities for stargazing. I learned that there was a big fire in King City. As some time went by it seemed like the cloud was moving east. Soon after Tom Piller arrived. We discussed the situation and agreed it might be okay since it looked like the cloud was starting to dissipate. 3 hours later the sky had cleared up and the SJAA Summer Deep Sky Scavenger Hunt was on.

I brought my Oberwerk 28x110mm binoculars

to take on the challenge of observing the intermediate list. The first two objects on the list were Centaurus A and Omega Centauri, these two wonderful southern hemisphere objects are always a challenge to see due to their low placement in the night sky. I found both of them just as the night sky went black. Omega Centauri did not disappoint, it's so big compared to M13 which most of us love to see. Other objects that I really enjoyed finding and viewing were M83, the Leo Triplet, M64, M23, M25, M68, East and West Veil Nebula. The most challenging to find were Collinder 464, La Superba, and 24 Comae Berenices. The conditions were getting better as the night got darker. We measured a SQM reading of 21.69 as the Milky Way was very visible as I was finishing up the list. Other notable objects not on the list were Collinder 316 so big and pretty, M24 Sagittarius Star Cloud through the binoculars, so beautiful! Stars filling the full frame with dark dust clouds visible through out the view. Very immersive 3-D type view!

I had made several rounds to check on others to see how they were doing. Everybody seemed to be enjoying the hunt through the lists. There were lots comments about how they were finding new objects they had never seen before, how much better the objects they usually looked at, looked tonight. Overall, it was a such a rewarding night for me seeing a renewed passion for our club members to really get and explore and learn the night sky and be enjoying the hobby. Let's keep it going and show some of the new objects you found at the next star party. Thanks again for everybody who made this event a great and rewarding event.

Marianne Damon

Message 1 of 9 , Jun 5, 7:07 PM

I can't thank Ed enough for organizing and hosting and providing coaching for this event. All of his prep work prepared me to arrive with confidence: travel directions, helpful hints, lists to be printed and copied into Sky Safari. As we set up, ashes drifted down onto our tarps and equipment! We were so fortunate that the rather thick smoke clouds cleared by viewing time (it reminded me of the clouds that finally cleared to give us one hour of viewing for our recent Mercury/Sun transit). I brought my planisphere, Messier and Caldwell Cards, Pocket Sky Atlas, my printed list with notes, and Sky Safari. By far, my most helpful aid was the list imported into Sky Safari. I worked from the Beginning List and found 21 of the 35 objects, plus, of course, Mars and Saturn, and 2 satellites that were fun to track. And we saw quite a few good meteors. I didn't get cold, didn't even put on gloves until 2 am. The camera-

derie was, as usual, wonderful, with people sharing views and helpful hints. This is the first time I've observed more than just a handful of objects, so this has really given a boost to my observing skills. Thanks so much!

Soni Kapoor

Message # of 9, Jun 5, 10:50 PM

What a wonderful night. All thanks to Ed for arranging, organizing and making sure on how everybody was doing during the event. I must say I was impressed by the list, I chose to complete intermediate list using both my binoculars and 8" Celestron SCT. It was little challenging at first not to use the goto feature of my CGEM mount, but became really fun after a while. I saw Omega Centauri for the first

time and I couldn't believe how big it was; almost 3 times larger than M13 in apparent size. As I continued on with the list, I saw a lot of new objects that I have never observed before and some were really tough to find. Thanks for the inspiration Ed!

Emmanuelle & Patrick

Message # of 9, Jun 6 8:03 PM

Ed - Thanks so much for setting up this fantastic event. We had a blast going through the advanced list, even if we could not finish it (most of the last objects were all rather faint large nebulae!). It was really fun to find new objects, and indeed the quality of the conditions was really great. Your lists are always so good! Thanks again, Emmanuelle & Patrick

David Balough

Message 9 of 9, Jun 9 12:29 AM

Again, thanks to Ed for taking the lead, I'm looking forward to more great nights of observing.

Four of us; Marianne, Phillip, Don and myself drowsed till daylight after observing till 4:30am, The Pinnacles were beautiful in the morning, we saw many meteors.

Editor's Note: This was one of the best nights we have had at Pinnacles West. Although I have heard recently that July 2, was even better. Barring smoky skies, there really doesn't seem to be any such thing as a bad night at Pinnacles West! Piller



Pinnacles 'Deep Sky Scavenger Hunt' participants

Photo Credit: Tom Piller

L-R: David Balough, Soni Kapoor, Carl Ching, Marianne Damon (kneeling), Alex Kramerov, Wolf Witt (kneeling), Emil, Julia Yang, Emil's friend, Phillip's Brother, Phillip Lieu, Ed Wong

Not pictured: Anand Raja, Tom Piller

SJAA Member Observation Posts

Saturn this morning

In-town planet observing report: June 10, 2016
From Vini Carter

I went into the back yard last night after the rest of the family had gone to bed. (As most of you know, our back yard is about a block away from Houge Park.) I wanted to get some more views of Mars, while it was still close. Mars turned out to be pretty boring. (It was even boring in SkySafari.) But Saturn was amazing. With the 8-inch Dob, the Cassini division was crisp. After midnight, when the mirror finally cooled down, surface markings of yellow and beige became stable, and moons Titan, Rhea, Tethys, and Dione were visible. (I had only seen Titan and Rhea before.) As seeing goes, it was not a great night. Moisture content was quite high, and only bright stars were visible but Saturn was great.

Message Response from Ed Wong June 10, 2016
Vini, Thanks for sharing your report with us. Glad to see you are enjoying the views through your scope. Nice to have a simple scope that you can just take out and go get some nice views without fussing with a lot of setup.

Pinnacles East Report - July 2016

From Natti & Bob

Hi Everyone - Just following up after our Pinnacles Group 128 campsite rental experience...It was AWESOME!

As it turned out, the surrounding group sites were never reserved, so Bob and I had the entire Group campsite valley area to ourselves.

There was plenty of wildlife to view during the daytime. Deer, bunnies, turkey vultures, and even a coyote the first night came down the hill behind us to pick up a road kill meal. It was the first time either of us had seen a coyote in person! The raccoon gang only had a single representative the first night, and after he realized there was no food, we never saw or heard from them again.

At night, the observing conditions were incredible! With no artificial light to be seen from the entire valley, we were able to walk around just by starlight. The first night was 1% moon, and the second was a sliver at 3%.

No campers stopped by and there were only a couple of cars after dark that drove through the upper road. Bob was able to get some imaging done later in the evening on the second night. With the batteries running low, he was still able to get an incredible picture of the Witch's Broom section of the Veil nebula with only (2) 5 min subs through the Red, Green and Blue filters.

The campground staff confirmed that midweek the group campsites are a ghost town. For \$75 per night, 10 people can camp out, or for \$110 up to 20. It was worth it for even the two of us to have the entire valley floor. Clear skies!



Bob's image of Witch's Broom, NGC6960, from Pinnacles East

Backyard Viewing

From: Vini Carter Jul 23, 2016

We were backyard observing last night, two blocks from Houge Park. It was one of the clearest and most stable viewing sessions we've had this year. (Ok, we haven't gone to Pinnacles yet this year.) Using 200x magnification on an 8 inch Dob, Jupiter was very clear (with three moons). Even in the far west and markings on the planet were clearly visible. But the real show was Saturn, with an amazing Cassini division and (I think) six moons [Titan, Thea, Tethys, Dione, Lapetus and Hyperion]. The "Teapot" was behind a tree and did not emerge until the moon rose.

Tonight, the moon won't rise until just before 11:00 (later at RCDO), so there will be more time (and no backyard trees) for viewing deep sky objects. Here's the Clear Sky forecast:

Waiting on Polaris at RCDO

6/25/2016



Photo Credit: Vini Carter

SJAA Member Observation Posts

Seeing Through the Fog From Mark McCarthy

The week of May 21 promised clear skies, so I was excited to get a little observing done. Astronomical twilight is not until after 9pm, and as these were work nights the sessions would be short. I used an 8-inch f/7.25 reflector to keep pace with the Astronomical League's Carbon Star and Stellar Evolution lists, which are well suited for this scope. However both nights were hampered by a thin marine layer blowing in. It was thin enough to see stars through, but it was illuminated as it passed in front of the waxing moon, like shower steam in front of a light bulb. The resulting wide spread glare washed out much of the naked eye stars, so my star hops were much longer than usual.

On one night the moon resided in Virgo, for example, and the only star I could make out in the whole constellation was Spica. I could still star hop to Porimma, and even split it, but I could not find my intended target SS Virginis in the glare, and needed to wait for another night when the moon was out of the way. All the same, both nights were enjoyable.

Here are the observations, all at 113x / 0.9°:

Y Canum Venaticorum, carbon star, "La Superba": Orange in finder, bright pure orange in the scope. By far the brightest in field, estimated magnitude 5.0, Morgan-Keenan C system* classification C5 [AAVSO 5.3, C5]. One of the reddest stars in the sky, the brightest example of J-star, a carbon star with large amounts of carbon-13. 19th century astronomer Angelo Secchi gave it its common name. It is likely in the final stages of fusing its remaining helium into carbon and shedding more material; it is ready to eject its outer layers to become a planetary nebula, with the remaining star to become a white dwarf.

RY Draconis, carbon star: Nice orange with a little yellow. Brightest in field, est. 7.5, C4 [AAVSO 7.2, C4]. Uncertain period of around 200 days, 6.0-8.0 magnitude range

SAO 157721 (Virgo), carbon star: Very faint yellow orange star at the end of string of two other stars. In glare of moonlight in thin marine layer. Est. 9.0, C4. [AAVSO N/A. 8.5, C2]

V Coronae Borealis, carbon star: Faint, small, faintly red in a rich field. Est. 9.5-10.0, C5 [AAVSO 10.0, C6]. Long period variable, from 6.9 - 12.6 over 357 days. To find this I used my 7x35 extra wide field binoculars to orientate myself, and I could fit the whole of this beautiful constellation in the field!

RR Herculis, carbon star: Small, faint, very red in arc of stars trailing NW to W. Est. 8.0, C5 [AAVSO 8.5, C5-C8]

SAO 46574 (Hercules), carbon star: yellow-red, brightest in field, in a string of four stars W-E. Est. 7.5-8.0, C2 [AAVSO N/A; 7.3 - 7.7, C3].

Beta Comae Berenices: Long hop from Eta Virginis due to poor transparency and moon halo. Very bright star, yellowish white color. Main sequence dwarf star. It is similar to our sun, slightly larger and brighter and only 3 billion years old.

Spica / Alpha Virginis: Very bright, blue-white star. It's a spectroscopic binary with orbital period of four DAYS! Each star is distorted by the other. Both rotate faster than their orbital period, and the highly elliptical orbit may mean this is a very young star system. It is a polarimetric variable: protostellar material might be entrained between the two stars.

Arcturus / Alpha Bootis: Very bright, yellowish. Brightest star in north celestial hemisphere. An orange giant, has exhausted the hydrogen in its core and in active hydrogen shell burning phase. In 1635 Jean-Baptiste

Morin observed it during the daytime with a telescope, first time anyone had observed a star during the day. A red giant star, 110x brighter than the sun, stronger in the infrared, while being only slightly larger than the sun. It has a high proper motion of two arc seconds a year, and a possible planetary system.

Theta Bootis: F-type star: Pretty, orange yellow color, bright. 47 light years from earth, so the photons entering my eye are the same age I am!

HD 139341, Bootes: Double star, both type K Blue-white A, slightly yellow B double star. 4/3 the mass of the sun but 11x larger, but only 3 billion years old. A planet discovered in 2009.

14 Herculis: Yellow-white, ordinary looking. It is an orange dwarf, 90% the mass, 70% the radius, and 36% the luminosity of the sun, but with 2x the metal content. Two planets have been discovered to orbit it.

SS Virginis, carbon star: SS Vir was blocked by moon earlier in the week, and is now out of the way; but still a long hop from Spica due to moon halo in marine layer. SS Vir is a faint orange in the finder and intense orange in the scope. Est. 7.0, C6. [AAVSO 6.8, C6]. It is a carbon Mira variable, hydrogen alpha emission increases as it grows brighter in its 361 day period. Radius is 500 sol.

T Draconis, carbon star: Had to pick up the scope and moved it to the other end of yard in order to sight this over my roof. Orange red and rather faint, rich field, Est. 8.0-8.5, C7. [AAVSO 10.0, C6-8].

* C0 is the lightest red and generally lower temperature, C7 the reddest and hottest. Estimating M-K is a requirement for the AL Carbon Star list, as well as visual magnitude since carbon stars are almost always variable. I reference the confirming values from AAVSO when possible.

SJAA Member Observation Posts

Monte Bello

From Alex Kramarov

Last night (July 30th), because the clouds/smog/haze forecast was "unclear" (pun intended), I decided to drive to Monte Bello since its only 30 minutes away from my home. I only bring the imaging gear on "good forecast" nights, so this trip was purely visual. The theme of the evening was to help my daughter "break in" her XT4.5 Orion Dobsonian (Dob), and I brought my 12" Lightbridge Dob.

We arrived around 8pm, quickly setup the gear, checked collimation, finder alignment (my daughter did great on the latter) and started searching for Venus. We were hoping to catch 5 planets, but Venus was only 8 degrees above the horizon when the sun set and we couldn't locate it. I was also hoping to catch Mercury since it was 4 degrees higher, but it also proved elusive. We then took a short walk to change our perspective and Mercury appeared. Apparently, from our previous position it had been obscured by a big tree on the hill to the west of us (and probably the same happened to Venus). The good thing about the XT4.5 that it is a grab and go, we quickly moved it 10 meters to the new position and caught the last few minutes Mercury was visible. It was so low that refraction was causing about 1 radius of Mercury to be filled in with blue on one side of the disk, and with red on the other side.

Moving to Jupiter, seeing was not great in that direction but we were able to see the 2 main cloud bands and 3 moons. Mars provided a much better view; 87% illuminated so presenting a nice "phase" display, ice cap visible at the "bottom", and Cyrtis major noticeable on the surface. Moving to Saturn we did see some graduated color banding close to the equator, Cassini division between the rings, and a few moons (Titan, Rhea, and the smaller Dione along with Tethys).

We interrupted our planetary observing to catch the ISS pass in the north with the 12" Reflector around 9:10PM. I

was tracking the station in the finder, and my daughter was looking through a 32mm eyepiece. We were able to determine that its "rectangular" shape is due to the layout of the solar panels.

At this point the Milky was became visible, which my daughter wanted to "explore". Since my preference was for her to find the objects herself, we started with the easy M4 (globular just 1.5 degrees west of Antares). M4 looked OK in her 4.5" Dob but was great with a 4mm eyepiece in the 12" Dob. We even spotted an orange star at the right of the viewing field of the cluster, and discussed that this is due to the low temperature of the star. Reading up later on the cluster, it seems that it has 2 distinct groups of stars in it, and many more orange/red stars (although we would probably need bigger aperture to see these, or darker skies). Lots of details and individual stars were visible, since it's a very close object to us (7.2k ly).

Next on the list was M80 (just between Antares and the top star of the Scorpius claw; easy to find just by pointing the red dot finder between the two). M80 is a very dense cluster, but appeared little more than a star with some nebulosity around it in the 4.5" and a bit better in the 12". M80 is a bigger cluster in terms of absolute numbers of stars but is much farther away than M4 (32,000 light years), so we couldn't see good detail in it.

At this point we jumped to a few objects that were visible with an unaided eye, so they could easily be pointed to with the red dot finders. M7 (Ptolemy cluster) was a large group of bright stars over a backdrop of a rich milky way area, even my 26mm 2" eyepiece left me wanting to get a wider view. M6 nearby was nothing to write home about though.

Next stop was a lengthy one; while my daughter was working to find M22, I spent time comparing M22 with M28 (M22 is much bigger and more impressive than M28). It was getting late so around 10:30pm I finished up with a run of M8 (Lagoon nebula), M20 (Trifid nebula), M21 (open star cluster, noth-

ing to write home about), M24 (an amazing, wide star cloud), M17 the Swan nebula (my daughter was not convinced that it is a swan), and M16 the Eagle nebula. I had to pull out my NPB filter to see M16. Going back to the swan with the NPB filter actually convinced my daughter, pretty easily, that M17 actually looks like a swan.

A note on this observing site: at the peak of the session there were more than 10 cars present and cars were coming and going throughout the evening, with lights and taking time to open and close the entry gate with its combination lock. While I like going to Monte Bello on weekdays due to its proximity, I cannot recommend going on weekends due to the "high traffic" of the fellow astronomers. If you go, while you may be tempted to "park" right in front of the entry gate for the best horizons the spot turns out to be the worst from the light interference perspective.

At 11pm we started packing up the equipment. Technically, this session was pretty interesting for the difference seen in the objects between the 4.5" and the 12" Dobsonians. Also, there is a significant difference in how easy it is to find deep sky objects with a 50mm finder (on the 12") and with the 26mm finder on the 4.5". While objects like M80 and M28 are on the edge of detection, for Monte Bello skies, the 50mm was making them easy to find. In the 26mm finder they were invisible.

Good session overall. On the way back down the hill we drove through the cloud layer and the sky disappeared in the orange glow reflected from the clouds.



SJAA Member Astrophoto Gallery

Subject: North America Nebula and the Pelican Nebula.

From: Bruce Braunstein

June 29, 2016

Location: Imaged in my back yard in San Jose.

Telescope: AT65EDQ

Mount: Atlas Pro

Camera: QHYCCD IC8300

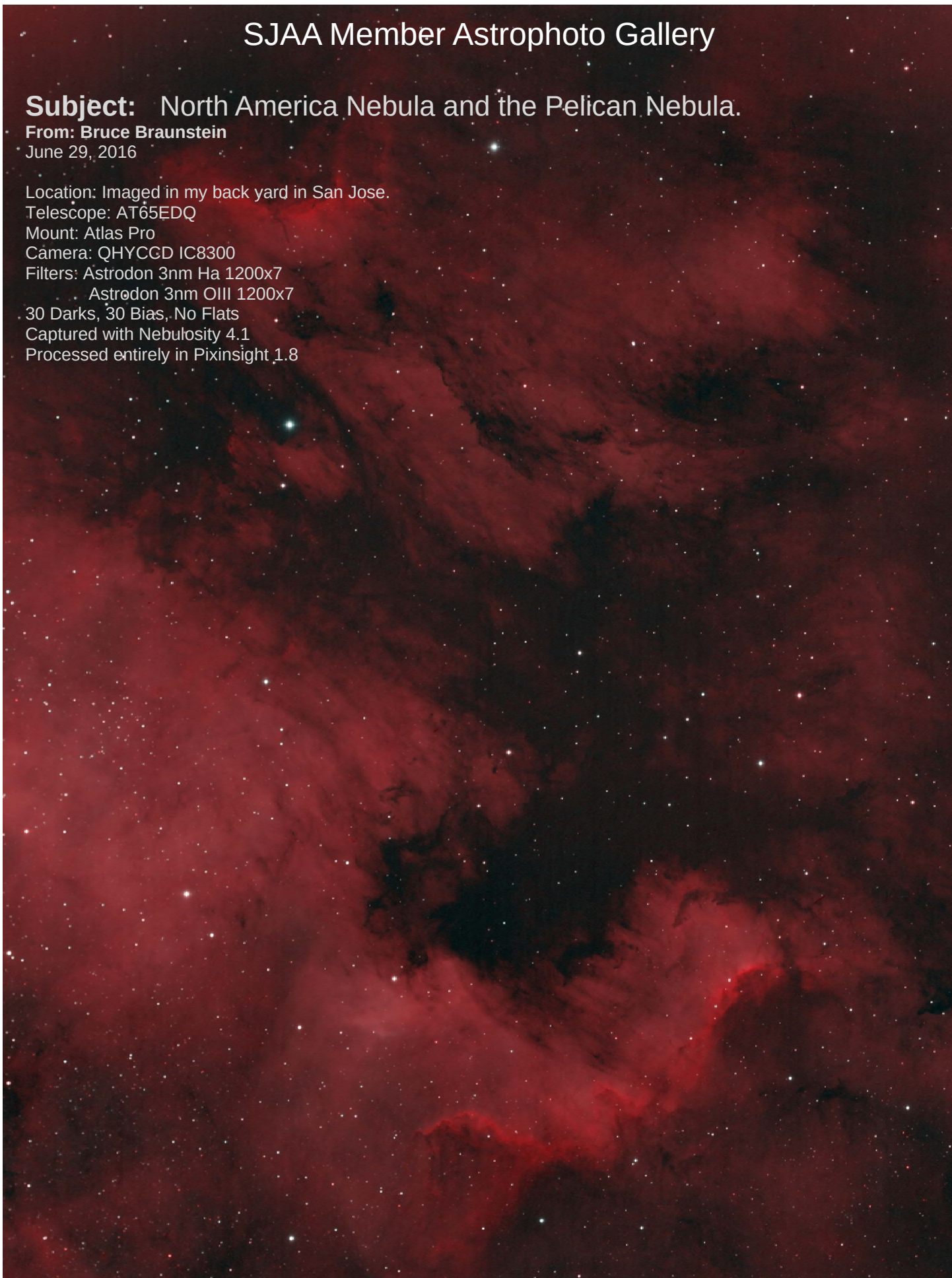
Filters: Astrodon 3nm Ha 1200x7

Astrodon 3nm OIII 1200x7

30 Darks, 30 Bias, No Flats

Captured with Nebulosity 4.1

Processed entirely in Pixinsight 1.8



Cave Nebula

From: Mark Striebeck

Location: Orange Spring Ranch, Mariposa, CA
Sept. 18, 2015

Imaging telescope: Takahashi TOA-130
Imaging camera: FLI MLx694
Guiding scope: Borg Mini 50mm
Filters: Astrodon Luminance, R,B,G
Darks: ~50, Flats: ~20, Bias: ~50

Mount: Astro-Physics Mach 1 GTO
Focal reducer: TOA-645 Super Reducer
Guide Camera: Starlight Xpress Lodestar
Integration: 5.4 hours
Processing Software: Pixinsight 1.8

Location: Ancient Bristlecone Pine Forest, CA
at 10,150 feet
July 7, 2016.

Canon EOS 6D (not modified) EF 16-35mm
f/4L IS USM at 35mm
f/5.6

Mount: iOptron
SkyTracker;
Exposure: (1) 60 second at ISO 6400; Canon
Digital Photo Professional 3.15 (raw)

Rich Field Milky Way above the 'Mexican Mine Cabins'
From Gary Chock

Iris Nebula, NGC 7023

Location: Coyote Lake County Park, Gilroy, California

Date: July 9, 2016

From: Swaroop Shere

Imaging telescope: Stellarvue 102 mm APO triplet
Imaging camera: QSI 583wsg
Mount: Orion Atlas EQ
Guiding camera: Starlight Express Lodestar X2
Focal reducer: Stellarvue 0.8X
Filters:
Astrodon Gen 2, 31mm Blue, Green, Red Filters
Software: PixInsight 1.8,
Integration: 3.5 hours
Flats: ~20, Bias: ~20



Milky Way

From: Bob Spangler

June 30th 2016, 10:20 pm

Location: Bryce Canyon National Park, Sunset point

Camera: Canon T6i; Rokinon 14mm 2.8 lens

Exposure: 22 second @ ISO 6400

M 106

From: PJ Mahany

February 9, 2016

Location: Yosemite, Ca

Telescope: ATM 6" F5 Newtonian

Field Corrector: Paracor II

Camera: Canon 60Da

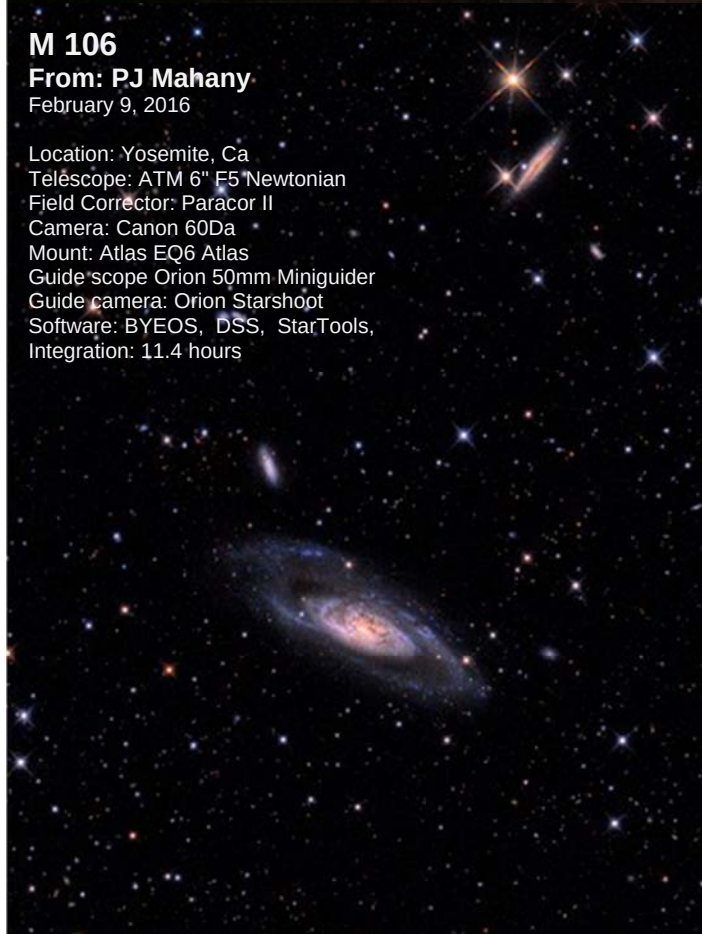
Mount: Atlas EQ6 Atlas

Guide scope Orion 50mm Miniguider

Guide camera: Orion Starshoot

Software: BYEOS, DSS, StarTools,

Integration: 11.4 hours



Collinder 316

From: Ed Wong

July 2, 2016

Location: Pinnacles West

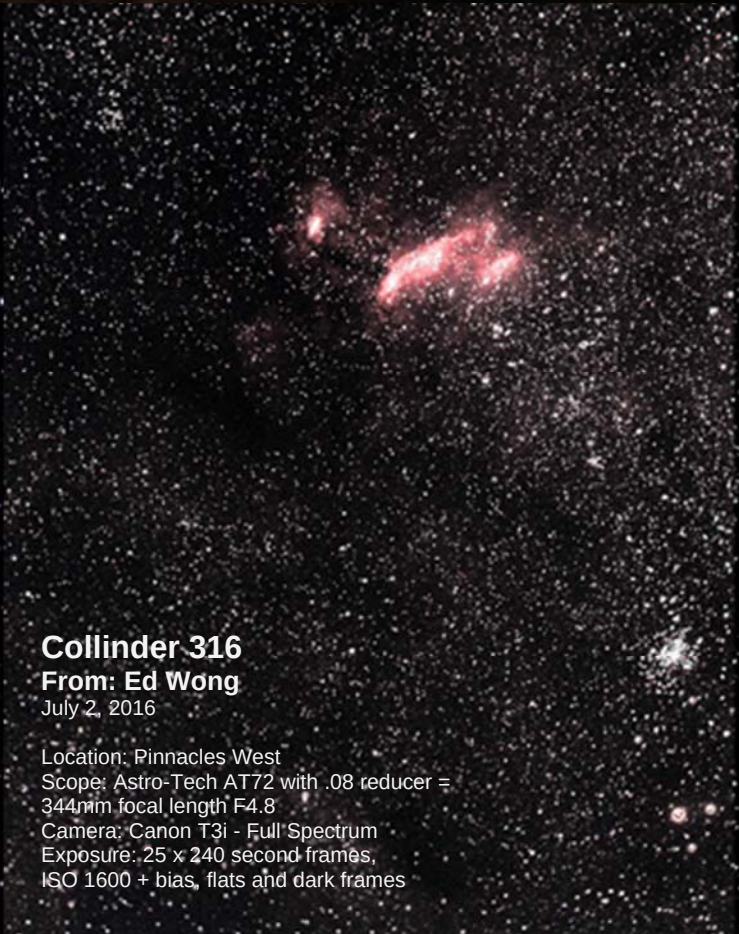
Scope: Astro-Tech AT72 with .08 reducer =

344mm focal length F4.8

Camera: Canon T3i - Full Spectrum

Exposure: 25 x 240 second frames,

ISO 1600 + bias, flats and dark frames



Eastern Veil - NGC 6995

By Manoj Koushik

July 30, 2016

Imaging dates: July 10, 15, 28 2016

Location: Manoj's new backyard observatory
"Tapas Observatory", San Jose, CA

Telescope: Stellarvue SVQ100
Imaging camera: QSI 683 wsg-8
Mount: Mesu 200
Filters: Astrodon 31mm - 3nm OIII;
3nm SII; 5nm Ha

Frames:
Integration: 44.0 hours
Darks: ~60, Flats: ~50, Bias: ~300
Processing: PixInsight version 1.8

Hi all, I've been away and quiet for a while with a variety of things not letting me do my favorite hobby. But meanwhile I have been whipping my home observatory into shape. Thoughts/suggestions welcome. This picture was shot from my backyard in San Jose. So, red skies basically....Manoj.

© Manoj Koushik



Kid Spot Jokes:

- ♦ **Why wouldn't you want to give Saturn a bath?**
Because he would leave a ring around the tub.
- ♦ **Why did the cow go to outer space?**
To visit the Milky Way.



Shane 120" reflector Lick Observatory
Photo Credit: Gary Chock

Kid Spot Quiz:

1. Name at least one of NASA's missions to:

- A. Mars
- B. Jupiter
- C. Saturn
- D. Pluto

Did You Know?

Source: Astronomy Magazine March 2015 issue "500 Coolest Things"

Earth is 3.1 million miles closer to the Sun at perihelion in early January than it is at aphelion in early July.

The full moon appears approximately 31' in diameter, or the size of a quarter at a distance of 8.85 feet.

1.5 million square miles. That's the area of Oceanus Procellarum, the Moon's largest feature.

British astronomer N.R. Pogson standardized the magnitude system in 1856.

The tallest mountain in the solar system - rising 14 miles from its base - is the central peak of the crater Rheasilvia on the asteroid Vesta. Rheasilvia formed as a result of a collision about 1 billion years ago.

Between 1645 and 1715, during a period now called the Maunder minimum, sunspots were rare.

El Gordo (ACT-CL J0102-4915) the largest known galaxy cluster, contains some 4.3 million billion times the mass of the Sun.

Of the 8,768 stars brighter than magnitude 6.5, only eight are brighter than magnitude 0.5.

Sagittarius the Archer contains 15 Messier deep-sky objects, more than any other constellation.

Kid Spot Quiz Answers:

- 1. Curiosity
- 2. Juno
- 3. Cassini
- 4. New Horizons

From the Board of Directors

Announcements

- ◇ The Calstar 2016 event is scheduled for 9/29 – 10/2, again at Whisper Canyon. Pre-registration is required.
- ◇ Saturday, September 10, 2016, from 10am - 3pm, SJAA will have Solar Telescopes out at the Coyote Valley Harvest Feast sponsored by the Open Space Authority (OSA). It is our biggest Solar event of the year and often draws up to 2,000 visitors. Details are posted on the SJAA meetup site at: <http://www.meetup.com/SJ-Astronomy/>
- ◇ The SJAA (half) Messier Marathon will be held in October so keep an eye out for the announcement.
- ◇ The SJAA bi-annual swap meet event will take place on Sunday, November 13th.
- ◇ There was a glitch with PayPal in July which affected membership renewals. A reminder email will be sent to all expired memberships that may have been affected by the Paypal outage.

Board Meeting Excerpts

May 21, 2016

In attendance

Dave Ittner, Rob Chapman, Bill O'Neil, Glenn Newell, Vini Carter, Lee Hoglan, Teruo Utsumi
Excused: Ed Wong, Rob Jaworski
Guests: Muditha Kanchana (member), Bruce Braunstein (member)

Google for Non Profits

Dave Ittner presented the status of [Google for Non Profits](#) (G4NPs) platform the SJAA Board and Directors group are migrating towards and walked through the basic set-up. The biggest issues are getting convenient access to multiple accounts and personalizing the name of the account owners. General information shared about storage of documents, photos and sharing between board members. Dave has offered classes on use of G4NPs, dates to be determined. Details and objectives to be posted on the SJAA Forum.

New Racks for the Storage Room

Dave Ittner proposed the purchase of a new rack system to better organize the storage room along with a Hogue Park Work day to reorganize the storage room. Bill O'Neil proposed that the HP Hall Work Day be coordinated with the storage rack

arrival day so the two events could be combined. To be discussed in more detail at the next meeting.

June 18, 2016

In attendance

Dave Ittner, Ed Wong, Vini Carter, Glenn Newell, Rob Jaworski, Bill O'Neil, Lee Hoglan, Rob Chapman
Excused: Teruo (Approved) proxy given to Dave
Guests: Bruce Braunstein, Wolf Witt, Muditha Kanchana

Google for Non Profits

Dave Ittner reported that the transition to G4NPs is moving forward and that he will provide a tutorial on June 19th at Hogue Park.

Open Board Seat

Ed Wong announced that he will retire from his SJAA board seat and Director position at the end of the current term. A search for Ed's replacement will begin immediately.

Storage Racks + HP Work Day

A formal proposal to purchase racks for the storage room was approved by all and Muditha Kanchana volunteered to pick them up. Dave Ittner will schedule a Hogue Park work day to install the racks and clean up the storage room.

Fix-It Day Tool Chest

Vini Carter noted that the equipment room could use some organization to help with the fix-it program. Also, per Ed Wong's recommendation, Vini will look into purchasing a fully supplied tool chest similar to what Ed used when he ran the program and then make a recommendation for purchase.

Lost & Found

Dave Ittner presented some options on how to handle items that are left behind. Dave noted that currently there is no designated place to store the 'found' items, there is no formal policy for lost and found and no way to announce what items are being held.

Rough outline procedure:

- Hold low value items for 6 months
- Hold high value items for 12 months
- Provide a list of the lost & found items on SJAA Meetup and the Club website.
- Sell any items at the swap meet that haven't been claimed (offer donation paperwork to anyone who asks after the sale)

To be discussed further at the next board meeting.

2017 Events Calendar

Dave Ittner asked that all Program leaders go to the SJAA Board Calendar and review

and approve all entries for 2017.

Lunt Double Stack

Bill O'Neil proposed that that a 'Double Stack' be added to the Club's Lunt Solar Scope along with a new battery for the solar scope tracking mount.. The double stack narrows the bandwidth of the band-pass solar filter from 0.7 Angstroms to 0.45 Angstroms. It produces better contrast on the solar surface especially for visual use. The motion to approve purchase of the new acquisition was approved by all.

City Council Meeting Report

Rob Jaworski reported that recently a San Jose City councilman proposed installing a pickleball court in the grass strip next to the sidewalk the SJAA uses for observing. Rob asked the City of San Jose to not take the grass strip away pointing out that the SJAA club uses the grass area during the In-Town Star parties. This discussion prompted more discussion on Club publicity and possibly making announcements using the City's events publication which has a distribution of approximately 7,000. Rob to follow up and report back.

July 16, 2016

In attendance

Dave Ittner, Ed Wong, Vini Carter, Glenn Newell, Lee Hoglan, Rob Chapman
Absent: Rob J (Excused), Bill O (Excused)

2017 SJAA Events Calendar

Dave Ittner presented the 2017 SJAA events calendar which was subsequently approved by all.

SJAA Forum

Dave Ittner gave an update on the SJAA Forum, smf.sjaa.net which is now live. Dave asked that board members use for Board discussions and provided a demonstration on use of the site.

Mendoza Ranch Permit Renewal

The Board approved a motion to renew the Mendoza Ranch annual site use permit with Santa Clara County Parks (SCCP)s. Ed Wong will follow through with SCCP .

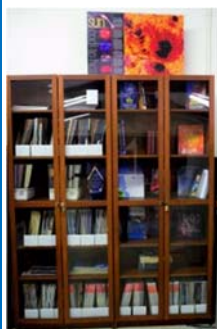
New Camera for Imaging Program

The board approved a motion to return the Club's CCD camera and purchase a \$2000 upgrade model which will better serve the Club's imaging program needs.

Online Security

Teruo reported that there have been some issues with unauthorized changes to events on the SJAA Meetup site and asked all present to please change their passwords.

SJAA Library!



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers!

Please check out our wish list on the SJAA webpage:

<http://www.sjaa.net/sjaa-library/>

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. The Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Bill O'Neil, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change:

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held every other month.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, each with a similar format, with

only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with a club telescope to get a better look at celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) has taken over the Loaner Program from Manoj Koushik. This program is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment below. Please note that certain items have restrictions or special conditions that must be met.

If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

(SIG), In February, Bruce Braunstein agreed to lead the SIG group. SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about. Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's Jim Van Nuland (Program Coordinator) for additional information.



<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President:	Dave Ittner
Vice President:	Ed Wong
Treasurer:	Rob Jaworski
Secretary:	Rob Chapman
Director:	Lee Hoglan
Director:	Bill O'Neil
Director:	Teruo Utsumi
Director:	Vini Carter
Director:	Glenn Newell
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Tom Piller
Fix-it Program:	Vini Carter
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Sukhada Palav
Loaner Program:	Muditha Kanchana
Memberships:	Anand Rajagopalan
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick START	Dave Ittner
Solar & Starry Nights:	Bill O'Neil
School Events:	Jim Van Nuland
Speakers:	Sukhada Palav
E-mails:	http://www.sjaa.net/contact

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles for publication should be submitted by not later than the 20th of the month of February, May, August and November. (earlier is better).

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

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San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type:

- ☐ **New** ☐ **Renewal** (Name only if no corrections)
- ☐ **\$20** Regular Membership with **online** Ephemeris
- ☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name: _____

Address: _____

City/ST/Zip: _____

Phone: _____

E-mail address: _____